

# HUMAN COMPUTER INTERACTION



# LECTURE CONTENT



- What is Human Computer Interaction?
- Goals of HCI
- Interface and its elements
- Interface types
- Physical and mental characteristics of a user associated with the perception, understanding and processing of interface
- User interface evaluation and its methods
- Prototyping

# **HUMAN COMPUTER INTERACTION**

Human-computer interaction (HCI) studies the design and use of computer technology, focused on the interfaces between people (users) and computers.

As a field of research, human-computer interaction is situated at the intersection of computer science, behavioural sciences, design, media studies, and several other fields of study. (Wikipedia)

# HCI GOALS



## THE MAIN PURPOSE OF HCI:

- improves the interaction between users and computers;
- ensures that computers are user-friendly, taking into account user needs.

## THE LONG-TERM GOAL OF HCI:

- development of systems that reduce the barrier between the cognitive model of what a person wants;
- understand and solve user tasks with the help of a computer.

# Factors that HCI designers need to consider first:

- what users want and expect considering their physical limitations and abilities;
- how information processing systems work, their accessabilities and availabilities;
- development of systems and programs that are pleasant and attractive to users;
- technical characteristics and limitations of computer hardware and software.



# INTERFACE

The user interface is an area of human-computer interaction (HCI)

An interface is a set of tools for the interaction of a user and a computer system in the broadest sense.

These tools are often

- buttons
- text boxes
- tips
- radio buttons
- drop-down lists, and more.



## INTERFACE ELEMENTS



- a set of user tasks that he solves using the system;
- the metaphor used by the system (for example, the desktop in MS Windows®);
- navigation between system units;
- visual design of software screens;
- tools for displaying information and formats; data entry devices and technologies;
- dialogs, interactions and transactions between the user and the computer;
- user feedback;
- decision-making on a specific topic; the order of program usage and the documentation for it.

# THE IMPORTANCE OF THE USER INTERFACE

- A well-structured interface and screen are very important for users because they understand the capabilities of the system through that interface.
- It is also a tool that offers many important tasks. These responsibilities often have a direct impact on the organization's relationship with customers and its profitability.
- The location and appearance of the screen affect a person differently. If they get convinced or confused, they will have great difficulties in their job and accordingly will make mistakes.
- Low-level design can affect users to escape from the system. It can also lead to tension, anger, and increases stress level.



# INTERFACE TYPES

## 1. COMMAND LINE INTERFACE

This is an interface type that allows users to interact with the computer's operating system by entering (typing) commands.

CLI is often used by programmers and system administrators, in engineering and scientific environments, as well as technically advanced users of personal computers.

The CLI is also popular with visually impaired people, as commands and responses can be displayed via updated Braille displays.

```
come to the ivanti platform command line for
anced configuration and triage capability.

a list of commands type help or ? followed
>?
filedirector - File Director commands
lookup       - Lookup host
ping         - Test connection through ICMP
restart      - Restart the system
shutdown     - Shutdown the system
logout       - Logout of the command line
help         - Get help about a command
shell        - Switch to shell
>ping dn-play-01
PING dn-play-01 (10.0.32.211): 56 data bytes
64 bytes from 10.0.32.211: icmp_seq=0 ttl=64
64 bytes from 10.0.32.211: icmp_seq=1 ttl=64
64 bytes from 10.0.32.211: icmp_seq=2 ttl=64
64 bytes from 10.0.32.211: icmp_seq=3 ttl=64
64 bytes from 10.0.32.211: icmp_seq=4 ttl=64

--- dn-play-01 ping statistics ---
5 packets transmitted, 5 packets received,
round-trip min/avg/max/stddev = 0.048/0.060/0.080/0.016
>
```

## 2. GRAPHICAL USER INTERFACE

Graphical User Interface (GUI) - this is a type of interface that allows users to communicate with operating systems in the form of non-text, i.e. images, animations and audio.

Users can use the mouse and keyboard to interact.

There are four GUI control groups:

- Input elements
- Output elements
- Selection items
- Action items



### 3. GESTURE BASED INTERFACE

The rapid development of technology was one of the reasons for the emergence of the gesture based interface. These include

- touch screens
- joysticks
- styluses and other elements.

This type of interface is widely used on the touch screen of modern smartphones (like iPhone) and laptops (with touch panel and touch screen) and other mobile devices.



## 4. VOICE USER INTERFACE

This type of interface, which uses a voice / speech, allows a person and a computer to interact, initiate, search for an automated service or process.

For example, saying "OK, Google" users interact with the voice interface of smartphone or tablet's operating system.



# Physical and mental characteristics of a user associated with the perception, understanding and processing of interface

## USER'S INFORMATION PERCEPTION

- Short term memory
- Long term memory

## USER'S COLOR PERCEPTION

- Warm and cold colors;
- Irritating colors on the interface

## USER'S SOUND PERCEPTION

- Attention sounds;
- Background sounds;
- Sounds as a source of additional information

## USER'S SUBJECTIVE PERCEPTION OF TIME

- User notifications;
- Time indicators with animated objects





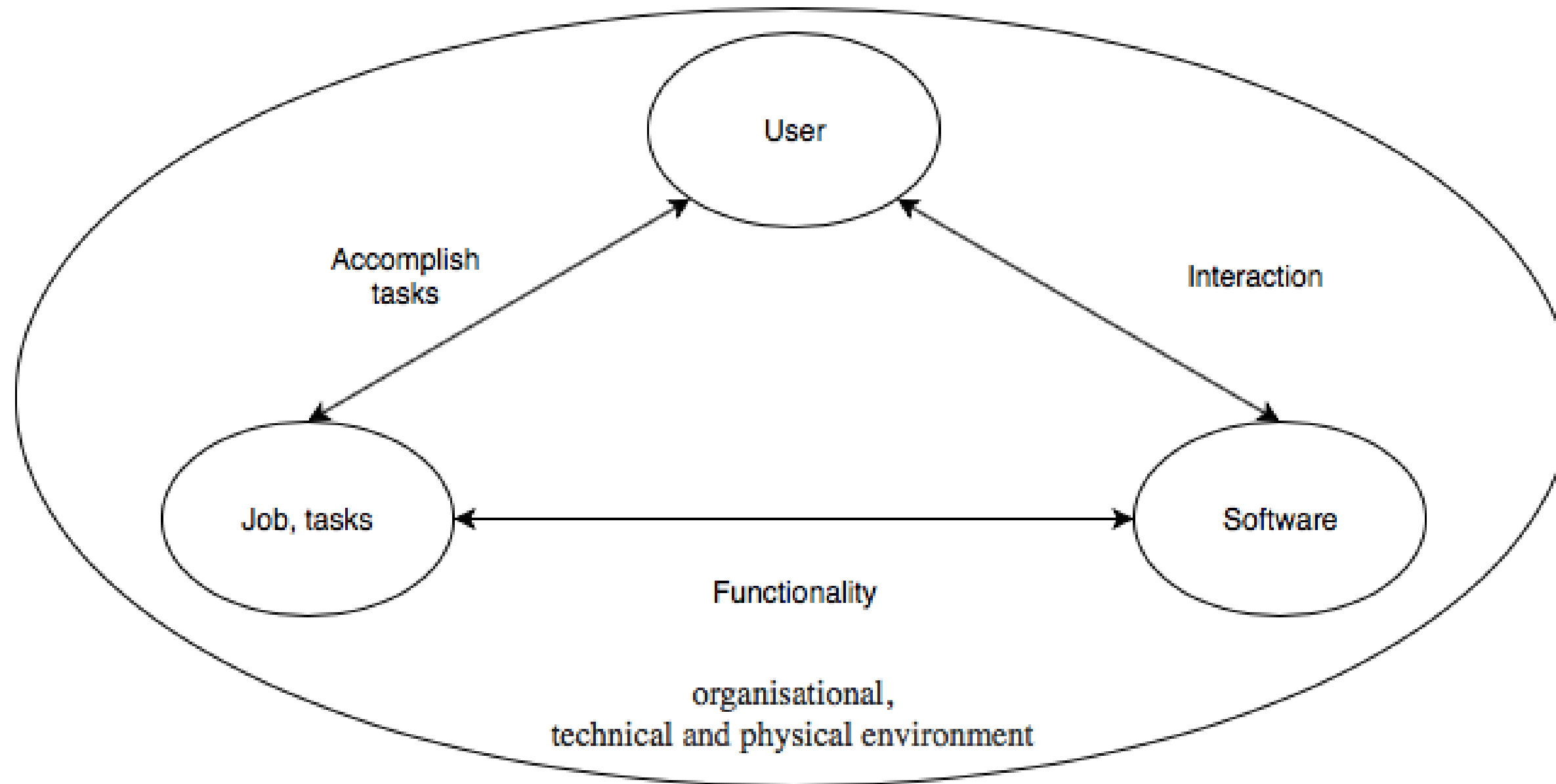
## USER INTERFACE EVALUATION

Special expert questionnaires, forms, checklists are used to evaluate the required level of interface usability.

An interface design that in many cases seemed easy to use for programmers may be difficult for the average user.

Therefore, in the absence of many interfaces, users are directly involved in the evaluation system to determine whether they are suitable or unsuitable for the general public before the software is introduced.

# ELEMENTS TO CONSIDER WHEN EVALUATING THE USER INTERFACE



# INTERFACE EVALUATION METHODS

## SUBJECTIVE EVALUATION METHOD

based directly on  
user feedback or  
judgement

## OBJECTIVE EVALUATION METHOD

paraexperimental  
studies;  
evaluation of system  
properties based on  
checklists;  
classical experiments.

## EXPERT BASED EVALUATION METHOD

the system is  
checked by an  
expert  
(objective +  
subjective  
evaluations)

## EXPERIMENTAL EVALUATION METHOD

involves comparing  
how different  
systems perform  
certain standardized  
tasks

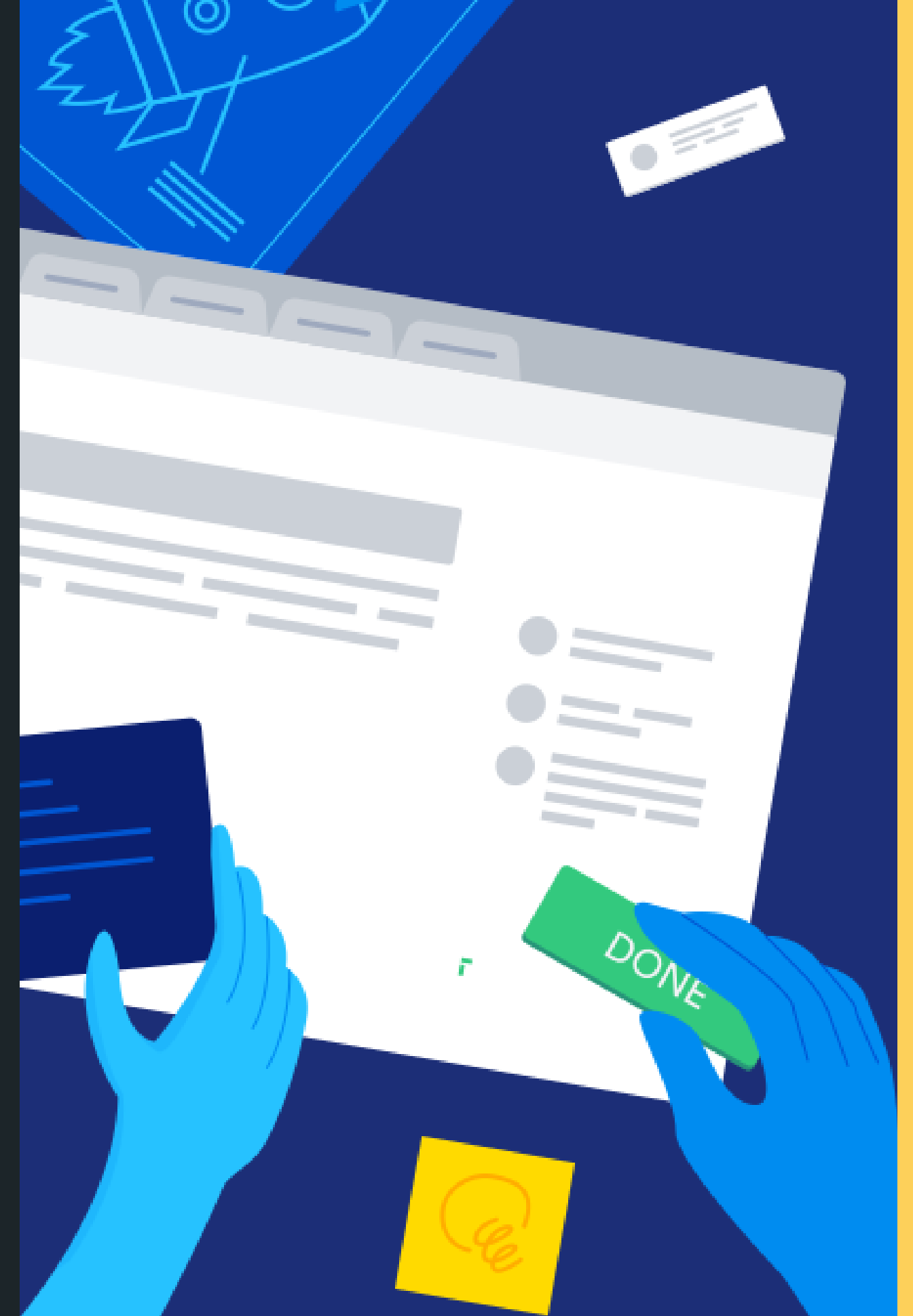


# PROTOTYPING

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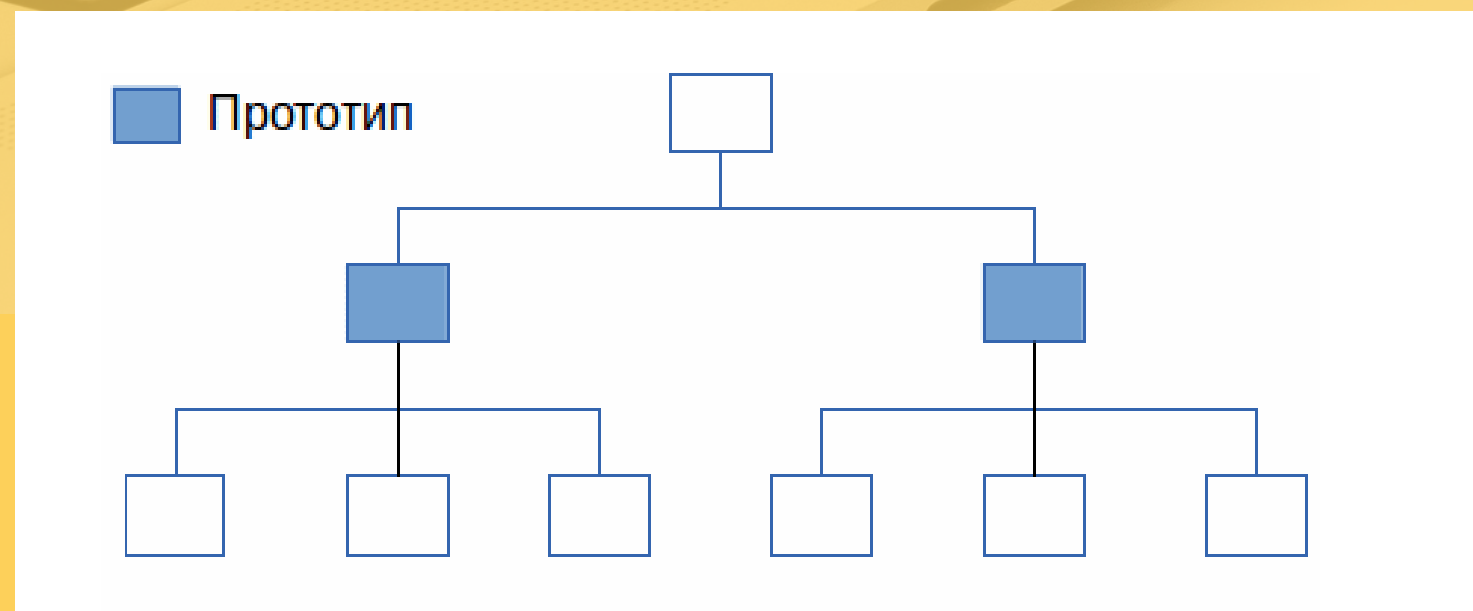
Prototyping:

- the stages of the interface or software development
- the process of creating a prototype of the program or
- the layout of the program (project, trial version)

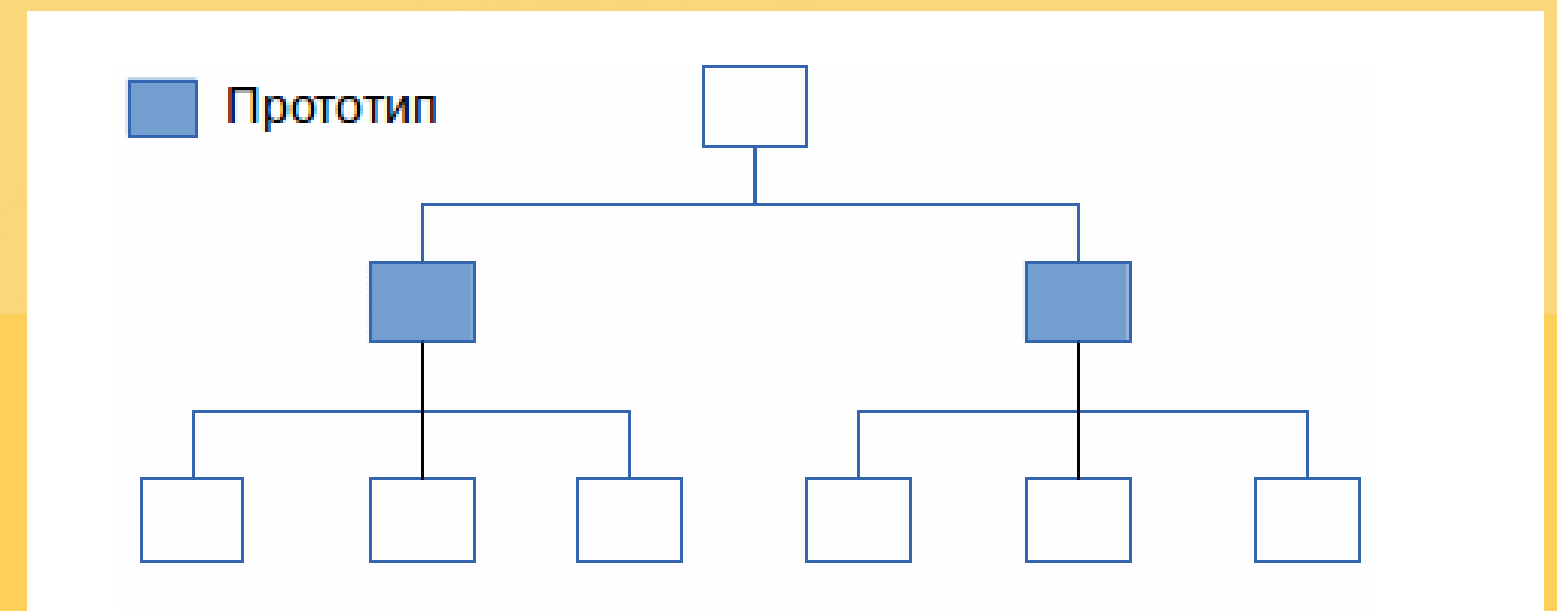


# PROTOTYPE SIZE

- Global prototypes model the whole system.
- Local prototypes model only a small part of the system. They can be used to resolve disagreements by comparing different design options. Two types of local prototypes, horizontal and vertical, are aimed at completing the range of functionality and capabilities in the prototype.



HORIZONTAL LOCAL  
PROTOTYPE

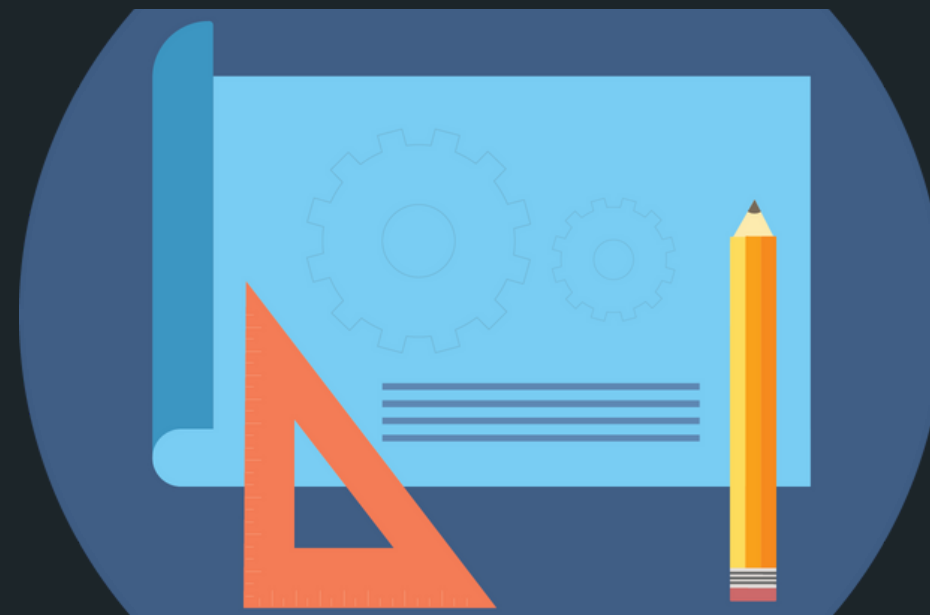


VERTICAL LOCAL  
PROTOTYPE

# Prototyping methods



EVOLUTIONARY  
PROTOTYPING



RAPID  
PROTOTYPING

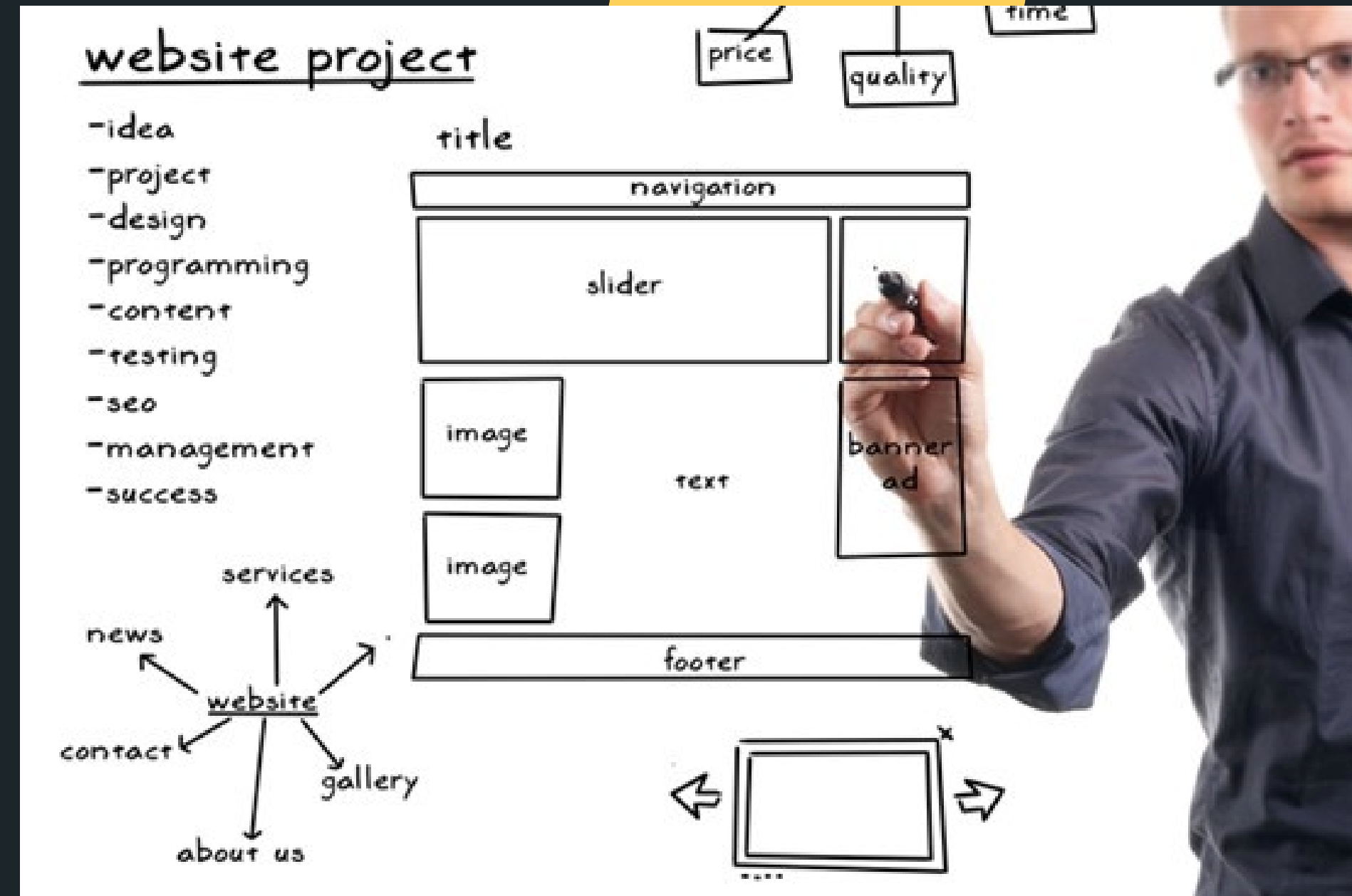


EXTREME  
PROTOTYPING

# PROTOTYPING

Prototyping is the fourth phase of both design thinking and design sprints. It's an essential part of user experience (UX) design that usually comes after ideation.

Prototyping can range in method and technique, which is often referred to as low, medium, or high fidelity. Fidelity refers to the level of detail and the functionality of prototype. Typically, the higher the fidelity, the more effort and cost it takes to create. Different scenarios call for different levels of fidelity, and there are pros and cons of each.





## DIFFERENT TYPES OF PROTOTYPING

- ***Low-fidelity prototypes***—often referred to as paper prototypes—are the quickest, lowest-effort prototypes one can create. They represent a simple, incomplete version of product, and are a great way to test high-level concepts prior to investing more time and energy in the design. Low-fidelity prototypes also allow to gather feedback and test concepts during the early stage of the design process.
- ***Medium-fidelity prototypes*** are the next level of prototyping. Often referred to as wireframes, medium-fidelity prototypes are digital and created in greyscale. They limit the design to user flows and information architecture while omitting branding elements, photos, and logos.
- ***High-fidelity prototypes***. After several rounds of testing with low- and medium-fidelity prototypes, it is time to invest time in rendering design in high fidelity. This is when the designer brings in branding, photos, copy, colors, and animations to bring their experience to life. As your design nears its final stages, it is important to test high-fidelity prototypes with users prior to the development phase.

# Prototyping tools



**ProtoPie**



**Sketch**



**Framer**

**proto.io**

**fluid**